

Architecture & delivery visual guide

Trading Desk v3 / Architecture, workflows, state ownership and phase gates

Research first. Deterministic risk. Human authority.
Paper-only remains a valid outcome through Phase 5.

02 Domains and shared controls

04 From approval to economic effect

06 Phases 4–5: certify and operate

08 Verification and acceptance

10 Order lifecycle and uncertainty

03 From evidence to approval

05 Phases 0–3: establish evidence

07 State ownership and memory

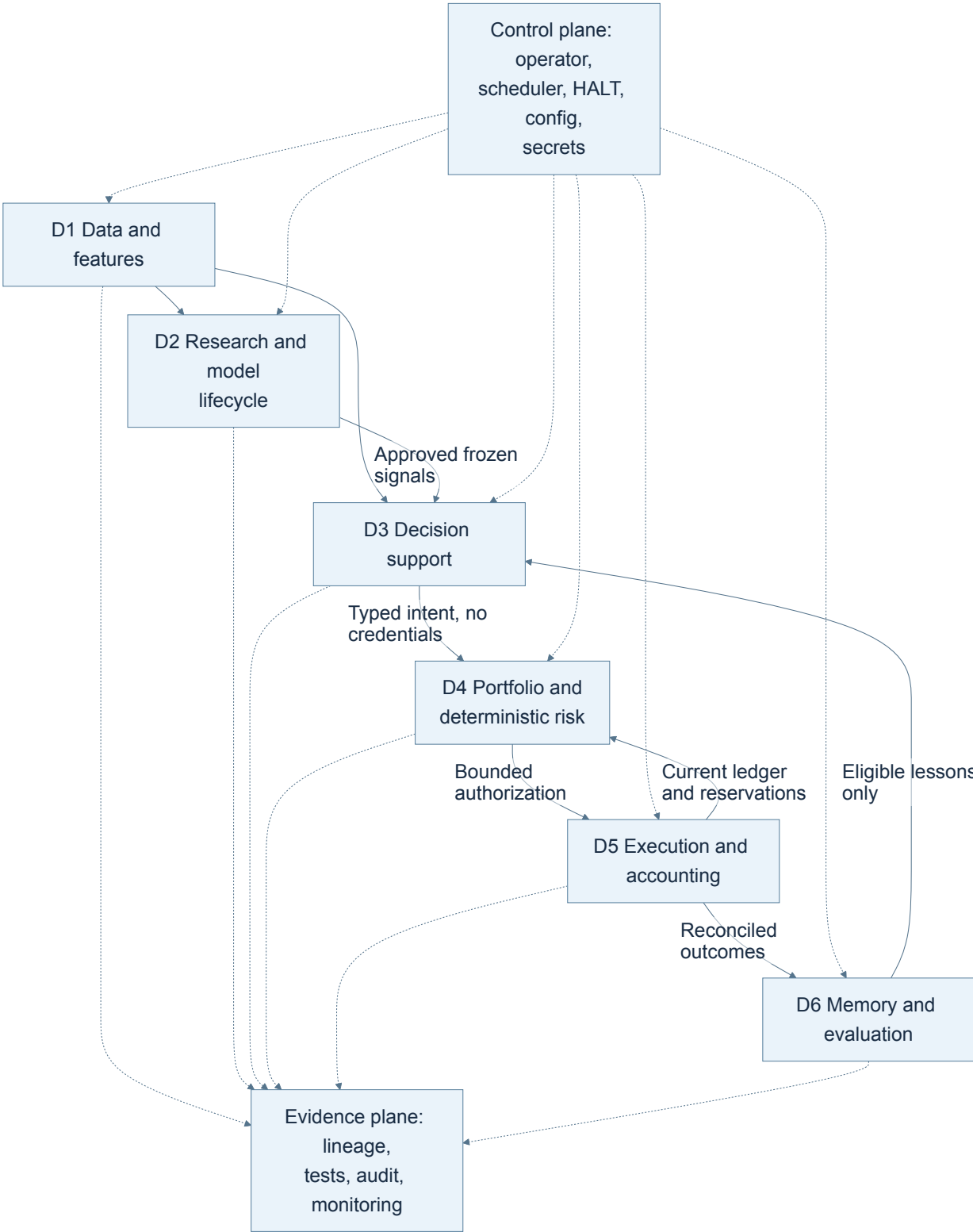
09 LangGraph workflow

11 Phase outputs and admission gates

Readiness: research and internal-paper prototype. Target diagrams describe the intended design, not implemented guarantees. No live broker adapter is provided.

Domains and shared controls

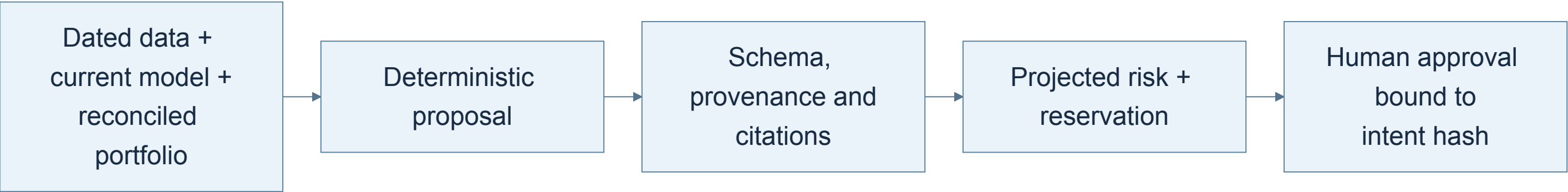
TARGET ARCHITECTURE / Six domains, one modular deployment



Control and evidence apply across every domain. LLM access is an optional adapter, not an authority layer. One Python package and one writer are the laptop baseline.

From evidence to approval

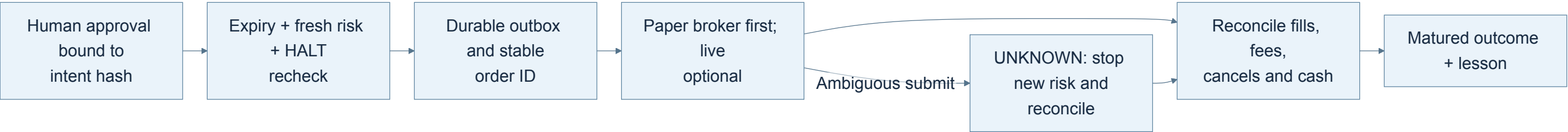
TARGET AUTHORITY FLOW / Part 1 of 2



Agents produce typed proposals, never broker orders. Approval is bound to an immutable intent; deterministic risk owns hard limits and projected reservations.

From approval to economic effect

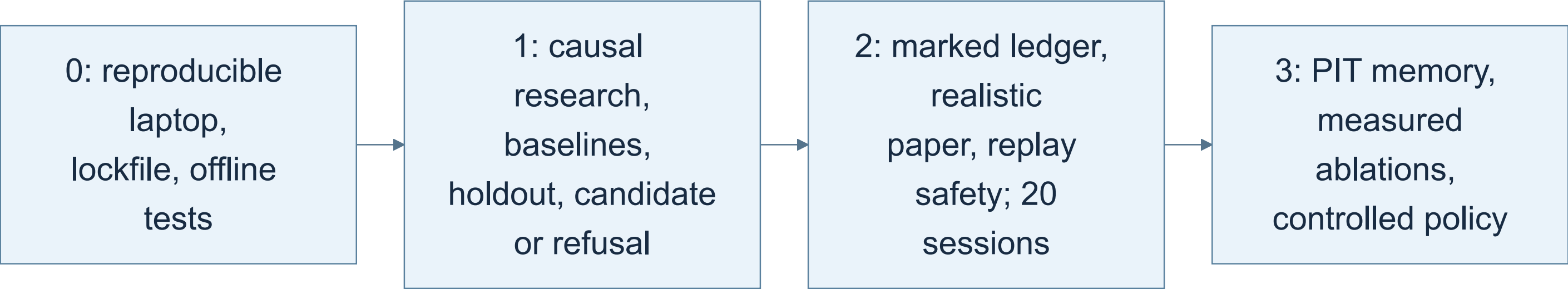
TARGET AUTHORITY FLOW / Part 2 of 2



Recheck expiry, current risk and HALT before dispatch. An ambiguous broker outcome becomes UNKNOWN and requires reconciliation, not a blind retry. These broker capabilities are not implemented in the starter.

Phases 0–3: establish evidence

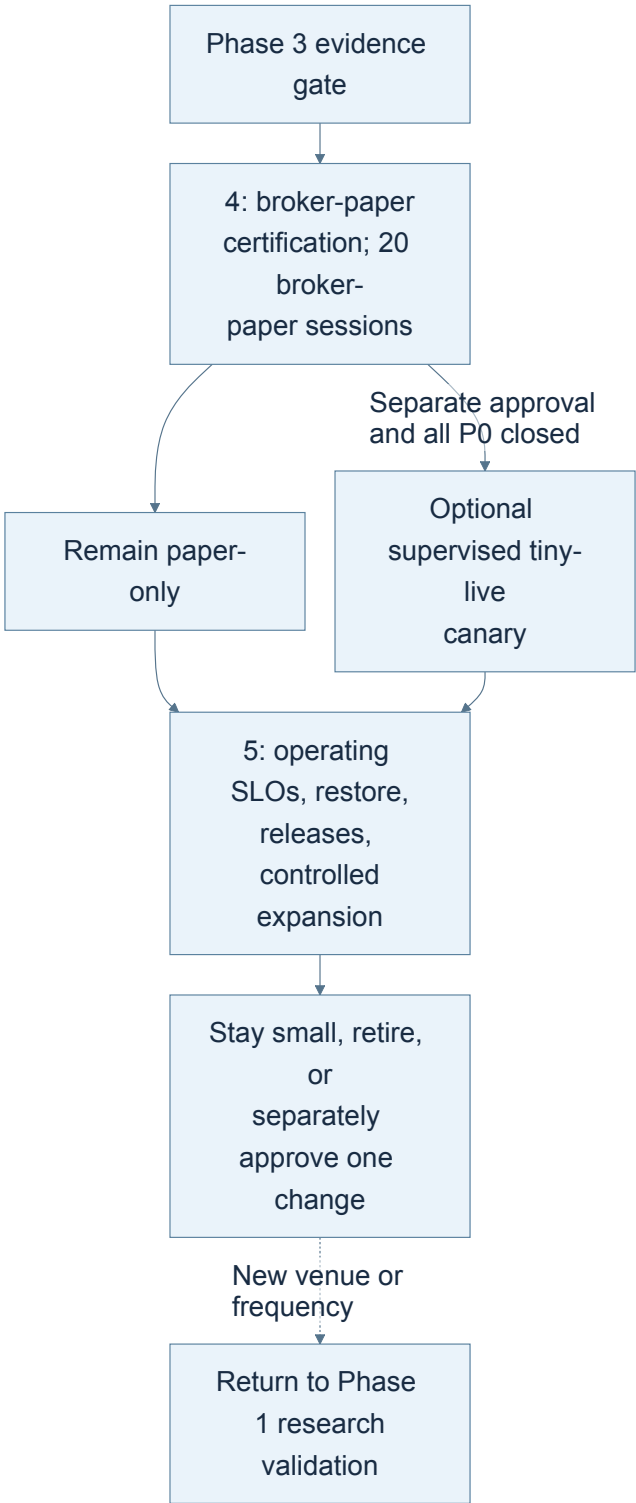
PROJECT ROADMAP / Foundations, research, paper lifecycle and measured memory



Phase 2 requires at least 20 prospective equity sessions after accounting and replay gates pass. A no-candidate research result permits simulator engineering, not live admission. Continue to Phase 4 only after the dependent evidence gates pass.

Phases 4–5: certify and operate

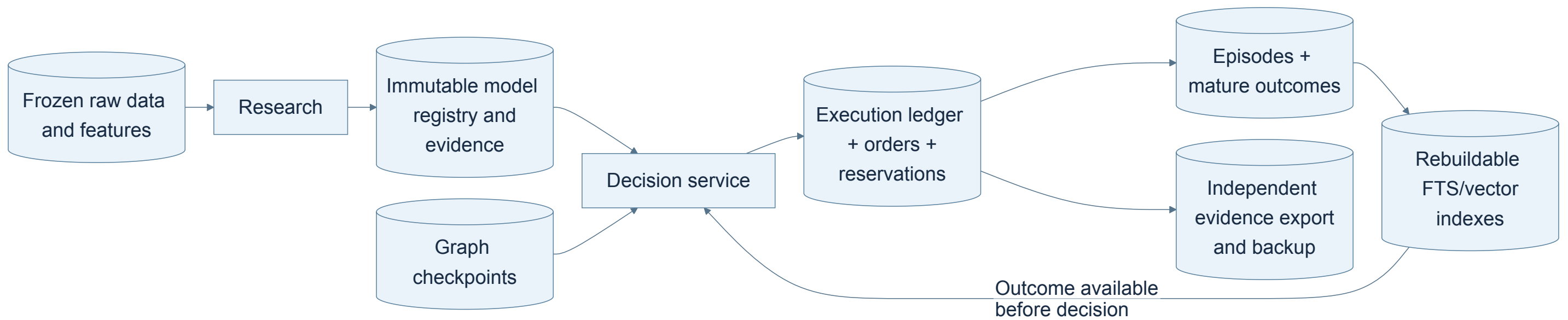
PROJECT ROADMAP / Broker paper, optional canary and governed operations



Observation windows are operating-policy floors, not proof of alpha or completion estimates. Phase 5 requires 30 calendar days of operational evidence and restore/rollback drills. P0 on the live branch means priority-zero blockers, not Phase 0.

State ownership and memory

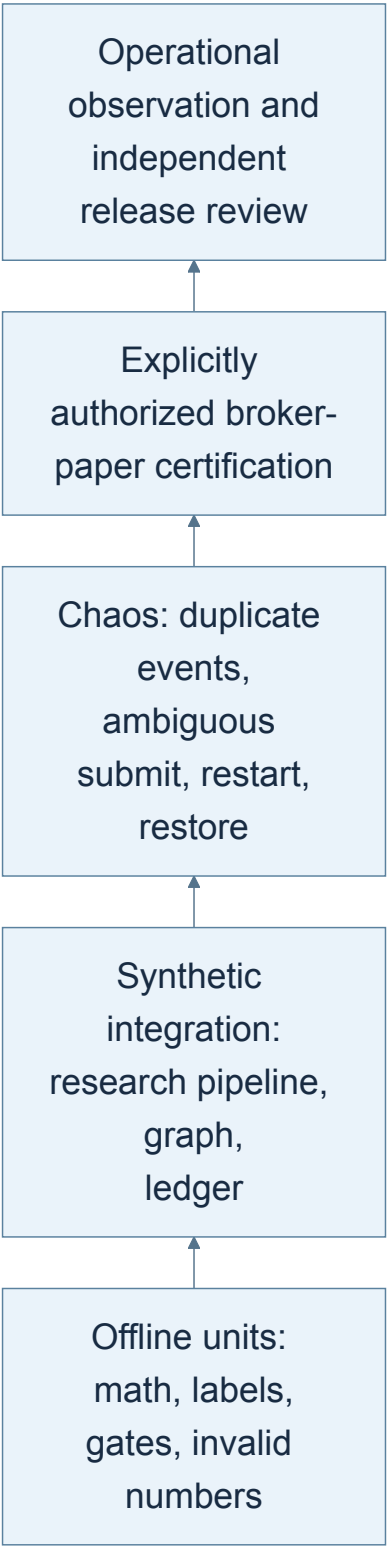
TARGET DATA RESPONSIBILITIES / Checkpoints are not portfolio truth



A restored checkpoint does not authorize a new economic effect. Preserve entry situations; retrieve only outcomes available before the decision. Indexes are rebuildable; the economic ledger is not a disposable cache.

Verification and acceptance

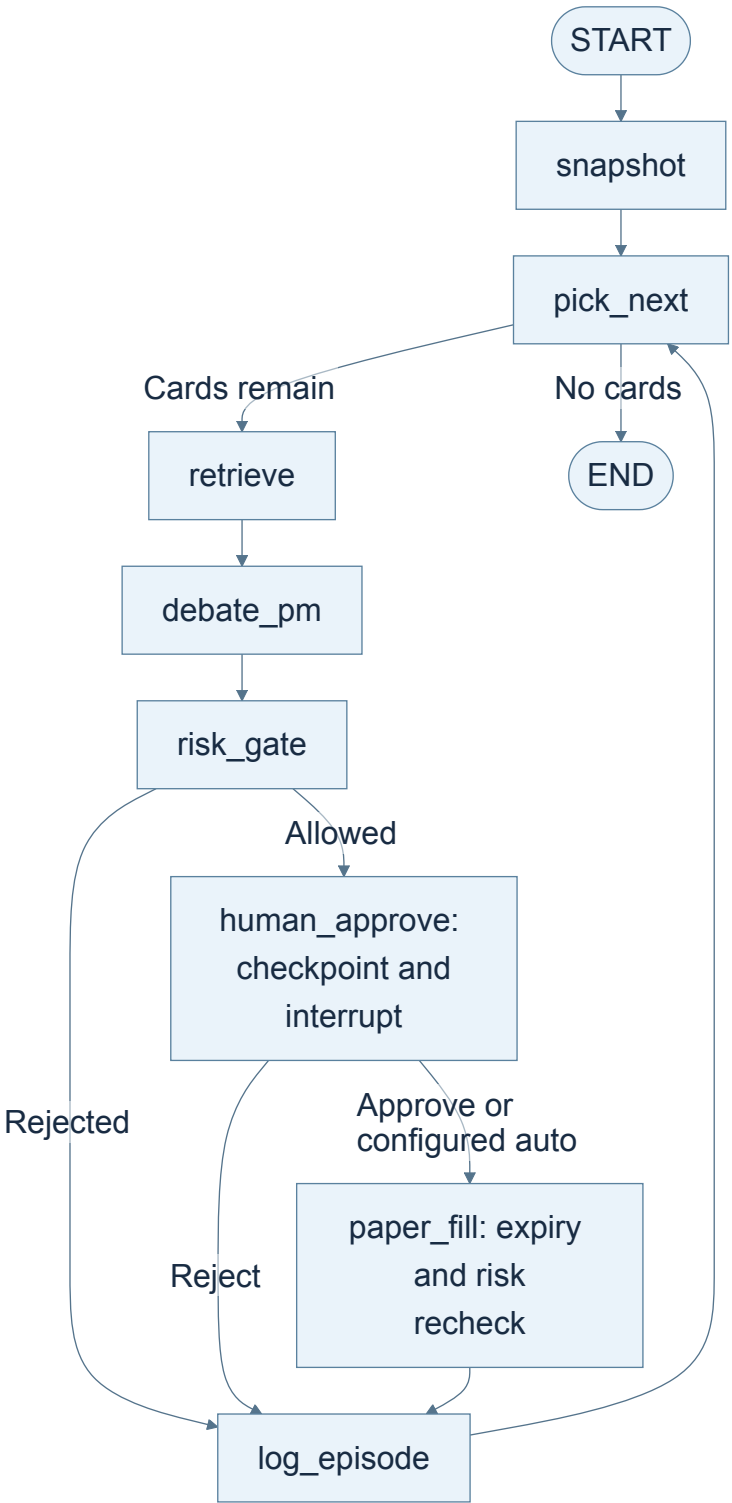
TARGET TEST PYRAMID / Evidence before external execution



The starter automates portions of unit and synthetic integration testing. Broker-paper certification, operational observation and independent release review remain separate gates.

LangGraph workflow

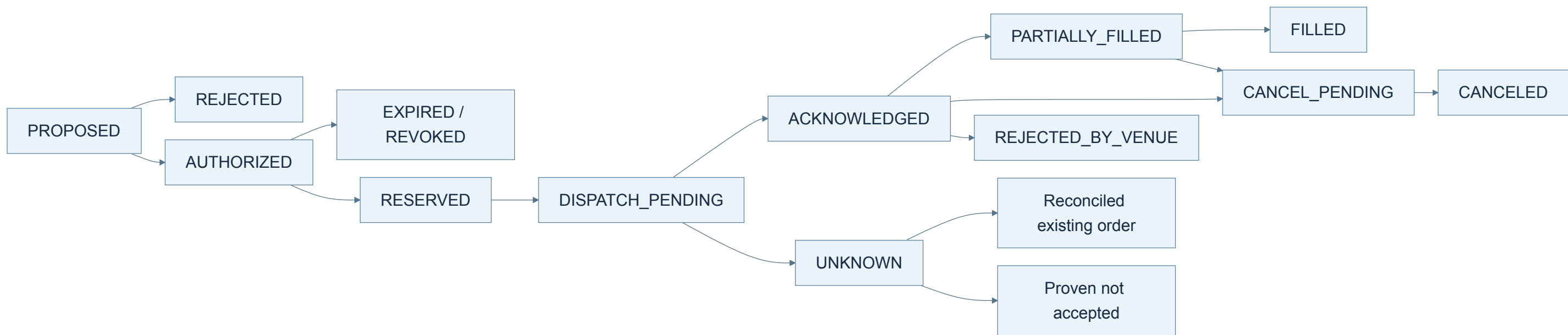
IMPLEMENTED PROTOTYPE / Internal paper only



A graph may pause for each allowed symbol. Approval expires after 300 seconds from the snapshot by default. Risk and HALT are rechecked; the local book still lacks production accounting and broker reconciliation.

Order lifecycle and uncertainty

TARGET EXECUTION CONTRACT / Not a shipped broker adapter



UNKNOWN is not rejected. Resolve acceptance through broker history and stable order IDs. A fill can race cancellation; reconcile filled quantity before releasing remaining reservations. A checkpoint is not an exactly-once transaction.

Phase outputs and admission gates

DELIVERY CONTRACT / Read each phase as an evidence gate, not a deadline

Phase	Ordered work packages	Output / admission gate
0 Reproducible laptop	0.1 tools; 0.2 locked environment; 0.3 isolated tests; 0.4 operating decisions	Clean install, green tests/lint, no accounts required, owner and budget recorded
1 Credible research	1.1 frozen data; 1.2 timing/labels; 1.3 baselines; 1.4 cost stress/holdout; 1.5 versioned model	Reproducible evidence, at least 3 valid folds where history permits, candidate or explicit refusal
2 Correct internal paper lifecycle	2.1 marked ledger; 2.2 order/exit state machine; 2.3 risk/reservations; 2.4 approval/replay; 2.5 paper observation	Accounting balances, negative tests and recovery drills; at least 20 equity sessions of prospective evidence
3 Measured memory and decision support	3.1 PIT lessons; 3.2 retrieval evaluation; 3.3 fixed/adaptive ablations; 3.4 drift/freshness; 3.5 optional LLM/policy	Frozen champion/challenger evidence; no increase in authority; disable additions without benefit
4 Broker-paper certification; optional tiny-live canary	4.1 separate adapter/account; 4.2 reconciliation; 4.3 fault injection; 4.4 broker-paper observation; 4.5 independent live decision	At least 20 broker-paper equity sessions, no unresolved safety incidents, venue evidence; optional supervised canary
5 Reliable operation and governed expansion	5.1 deploy/recover; 5.2 SLOs/budget; 5.3 release/rollback; 5.4 champion/challenger; 5.5 scope review	30 calendar days of operational evidence plus successful restore/reconciliation drill; paper-only completion is valid

Engineering completion and model admission are different. A no-candidate result allows simulator development, not live trading. Any unresolved priority-zero blocker prevents the dependent external-execution capability.